上海创智学院
Shanghai Innovation Institute

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Reinforcement Fine-Tuning LLMs

Roselia: Robot Code Policy RFT Is Sensitive to Trajectory Quality

Base-policy GRPO and reward-filtered teacher warm starts

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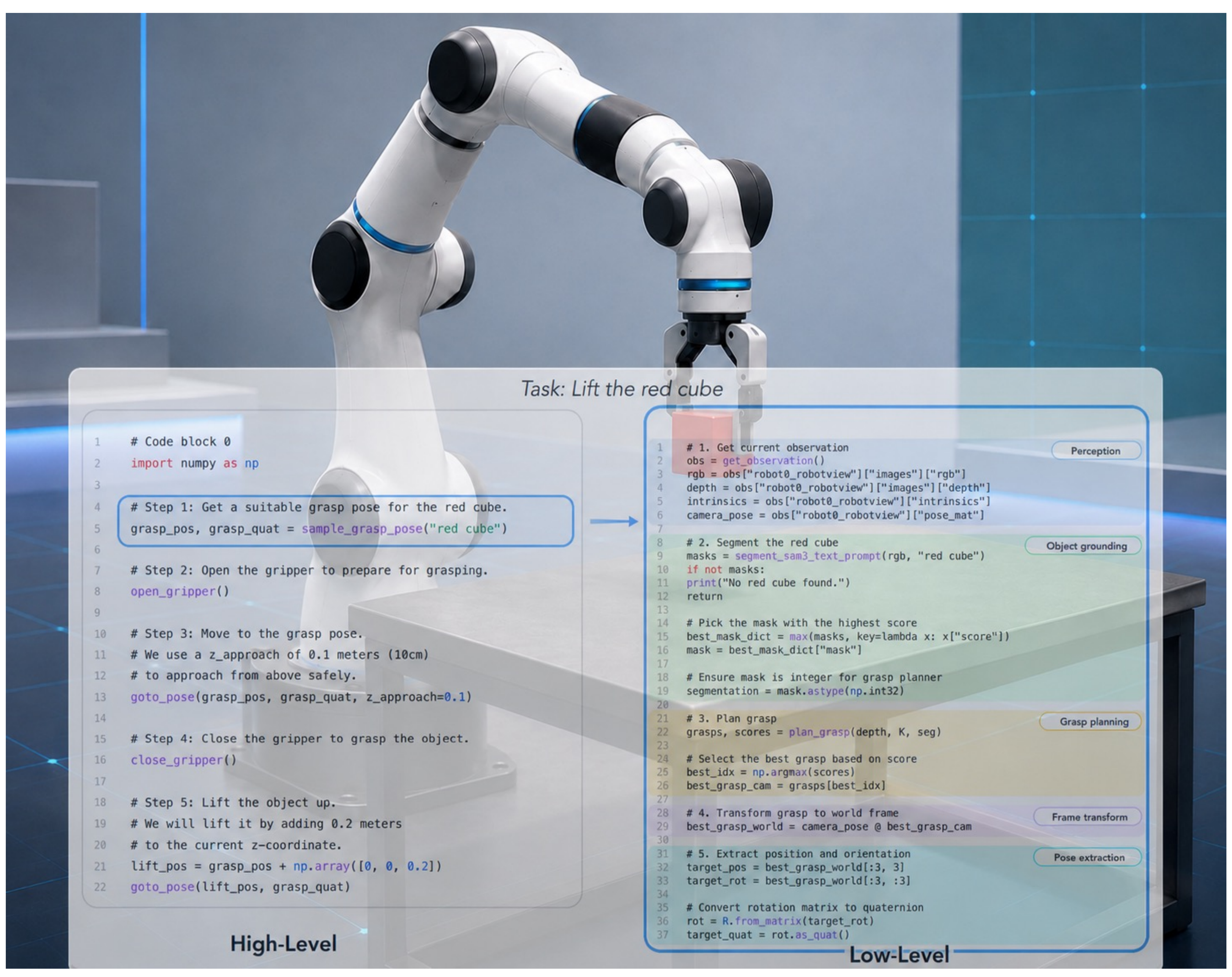
Code Repo

Code-as-Policy Robot Control

Code-as-Policy uses an LLM to generate executable robot programs from task instructions and environment observations. Instead of outputting low-level actions directly, the model writes structured control code that calls robot APIs, motion primitives, and perception utilities. This makes policies more interpretable and easier to verify, but also makes success highly sensitive to code format, execution errors, and the quality of generated trajectories.



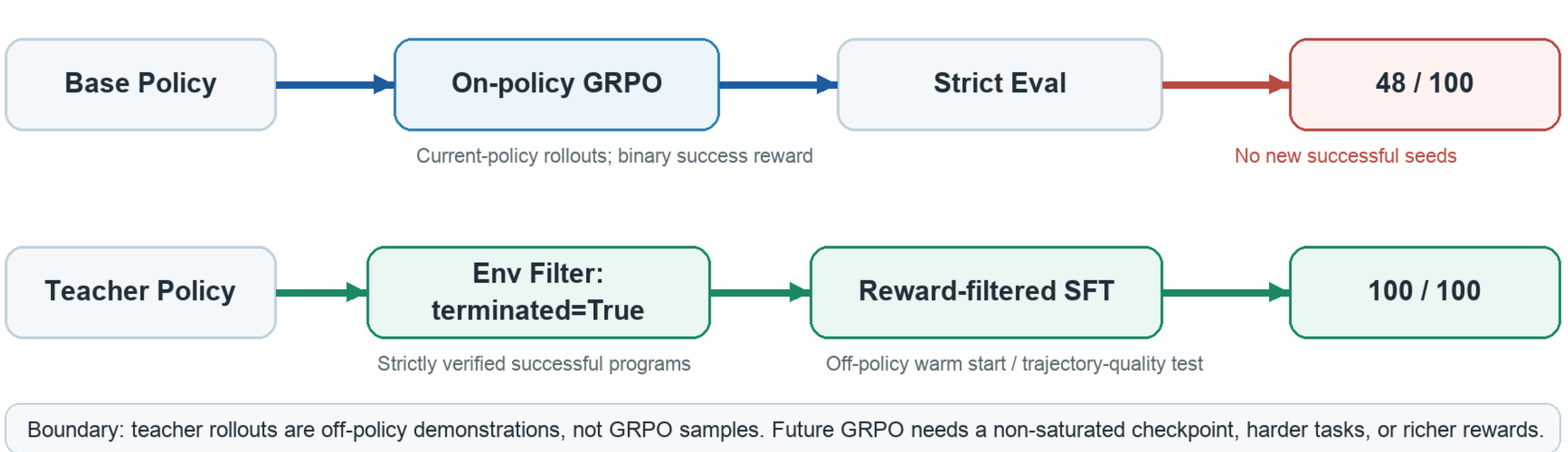
Motivation

Can reinforcement fine-tuning improve an open-source code-policy model for robot manipulation? We study why direct on-policy GRPO from a weak base policy fails, and whether rollout trajectory quality is the limiting factor.

Core Hypothesis

For robot code-policy RFT, the rollout policy determines the quality of learning signal. Weak base-policy rollouts may contain sparse and noisy successful trajectories, while stronger teacher rollouts can provide a better trajectory distribution for the same student model.

Method Overview



Fair Evaluation

Student_model: Qwen2.5-Coder-7B-Instruct

Metric: Task_strict_success, simulation terminated=True

Shaped reward is used only as a diagnostic. A rollout is counted as successful only when the simulator reports terminated=True.

Main Result

Same task, prompt, parser, decoding, evaluator; success = terminated=True

Method	Rollout / Data Source	Strict Success	Interpretation
Base model	None	49 / 100	Main baseline
Base-policy GRPO	On-policy base/student rollouts	48 / 100	No improvement; 0 GRPO-only successes
Base self-rollout data	Base sampled candidates	174 / 512	Sparse successful candidates
Teacher-SFT pilot	128 verified teacher successes	100 / 100	+51 points over base
Teacher-SFT large	2048 verified teacher successes	100 / 100	Same student; teacher not used at test time

Takeaway Teacher-verified trajectories transfer to the same open-source student, while base-policy GRPO does not improve strict success.

Diagnosis

Direct base-policy GRPO did not improve strict task success. The medium GRPO checkpoint solved 48 seeds, all of which overlapped with base-model successes. It produced 0 GRPO-only successful seeds. In contrast, strictly verified teacher trajectories transferred effectively into the same open-source student model.

Why Not More GRPO?

The Teacher-SFT checkpoint saturates the current benchmark at 100/100. With a binary success reward, GRPO has little gradient signal when rollout groups are all successful. Meaningful follow-up RL requires a harder task distribution, richer success-conditioned rewards, or a non-saturated warm-start checkpoint.

Conclusion

Direct on-policy GRPO from the base policy failed to improve strict task success. Reward-filtered teacher trajectories improved the same student from 49/100 to 100/100.

This indicates that trajectory quality and reward/evaluator alignment are key bottlenecks for RFT in robot code-policy settings.

Future works

Apply teacher-guided SFT on harder tasks where the warm-start student is **not yet saturated**. A non-100% checkpoint can then initialize on-policy GRPO, using student-generated rollouts and strict terminated=True rewards to improve beyond behavior cloning.

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